BASE CHANGE STABLE PROPERTIES TRANSFER TO THE RESTRICTIONS OF A GROUPOID

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The defining diagram of a restriction, exhibiting s' as a base change of h and relating t' to s' (more-groupoids chapter, equation-restriction, carried as context span 899-918). cited at proof lines 973-974. Exhibits s' as the base change of h, giving s' has P, and the symmetry giving t' isomorphic to s'.

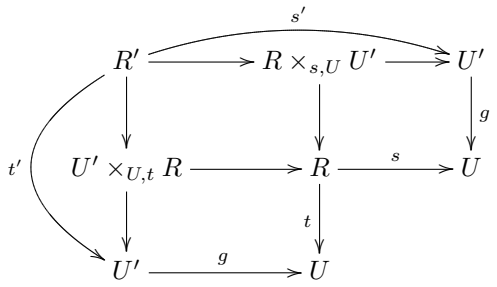
Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. Let $g : U' \to U$ be a morphism of schemes. Let (U', R', s', t', c') be the restriction of (U, R, s, t, c) via g, and let $h = s \circ pr_1 : U' \times_{g,U,t} R \to U$. If $\mathcal{P}$ is a property of morphisms of schemes such that*

(1) *h has property $\mathcal{P}$, and*
(2) *$\mathcal{P}$ is preserved under base change,*

then s', t' have property $\mathcal{P}$.

In this section we collect a bunch of lemmas on properties of groupoids which are inherited by restrictions. Most of these lemmas can be proved by contemplating the defining diagram

(0.2.1)



Proof. This is clear as s' is the base change of h by Diagram (0.2.1) and t' is isomorphic to s' as a morphism of schemes. □

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